

Anora - A Mental Wellness App

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Abstract Overview

Anora is a privacy-first mental wellness journaling application that leverages on-device AI to provide clinical-grade insights without compromising user data. By processing all sensitive text locally and using federated learning for model improvement, Anora bridges the gap between personal self-reflection and professional clinical care, offering DSM-5 aligned indicators while ensuring zero-knowledge privacy.

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Problem Statement

PRIVACY-UTILITY TRADE-OFF

Current digital health ecosystems force users to sacrifice privacy for utility; users must upload intimate reflections to central servers to receive data-driven insights, creating a "Privacy Paradox".

CENTRALIZED VULNERABILITY

Traditional cloud architectures create "honey pots" of Protected Health Information (PHI). These centralized databases are susceptible to mass data breaches, interception, and monetization ("surveillance capitalism").

TRUST DEFICIT

There is a critical absence of "Zero-Knowledge" tools in the market that simultaneously offer AI-driven DSM-5 risk detection and secure, verifiable data sharing mechanisms for clinical intervention.

Background

DOMAIN CONTEXT

The intersection of Mental Health and AI is growing, but existing solutions largely fail to respect Data Sovereignty. Most systems effectively transfer ownership of the user's intimate reflections to the service provider upon upload.

VULNERABLE POPULATIONS

For high-risk demographics—such as victims of domestic abuse or individuals in restrictive environments—the threat of data interception acts as a barrier to seeking help. A system that leaves a data trail on a server is fundamentally unsafe for these users.

TECHNICAL PRECEDENTS

While BERT-based models offer state-of-the-art performance for text classification, their computational cost has historically mandated server-side inference. Anora addresses this by applying quantization techniques to deploy these models on resource-constrained mobile devices.



Gap Analysis: Problems and Advancements

LIMITATIONS OF STATE-OF-THE-ART

- **Cloud Dependency:** Most competitors require active internet connections for AI functionality, introducing latency and service unavailability in offline scenarios.
- **Trust-Based Privacy:** Current apps rely on policy-based privacy ("We promise not to look"). This is inferior to architecture-based privacy ("We cannot look"), which Anora provides.

PROPOSED SOLUTIONS

- **Edge Intelligence:** By moving the inference engine to the client, Anora eliminates the network as a point of failure and a vector for attack.
- **Verifiable Secrecy:** The use of Hybrid Encryption (RSA+AES) for the "Clinician Locked Box" creates a verifiable cryptographic channel that bypasses the platform administrator entirely.



Aim & Objectives

PROJECT AIM

To engineer a mobile application capable of delivering clinical-grade (DSM-5 aligned) mental health tracking and secure clinician reporting within a strict Zero-Knowledge privacy framework.

TECHNICAL OBJECTIVES

- Encrypted Persistence: Implementation of AES-256-GCM encryption for all local SQLite/Hive database entries to secure data at rest against physical device compromise.
- On-Device NLP: Integration of a quantized MentalBERT model via the TFLite interpreter to achieve real-time (<200ms) emotion classification and risk flagging locally.
- Cryptographic Reporting: Development of the "Clinician Locked Box" module using asymmetric cryptography (RSA-2048) to facilitate secure, user-controlled data sharing.
- Privacy-Preserving Training: Deployment of a Federated Learning client to execute local training loops and participate in Secure Aggregation protocols for global model updates.

Proposed Solution

THE "BLIND MAILMAN" ARCHITECTURE

The backend infrastructure is designed with a "Least Privilege" philosophy. It acts solely as a router for encrypted blobs. It does not store user passwords or encryption keys, rendering it blind to the content it transports.

LOCAL-FIRST ECOSYSTEM

The mobile application is the primary source of truth. Data ingestion, tokenization, inference, and encryption all occur within the "Trusted Zone" of the user's device before any network request is even constructed.

CLEAN AI TRAINING

By utilizing Secure Aggregation, the system mathematically ensures that the central server can only see the sum of model updates from thousands of users, making it impossible to reverse-engineer any single user's contribution.



System Architecture Overview

TRUSTED ZONE (USER DEVICE)

- **Secure Enclave:** Leveraging hardware-backed security modules (iOS Secure Enclave / Android Keystore) to manage encryption keys, ensuring they are never exposed to the application layer in memory.
- **Inference Engine:** A dedicated TFLite thread capable of running the quantized MentalBERT model on the device CPU/NPU.

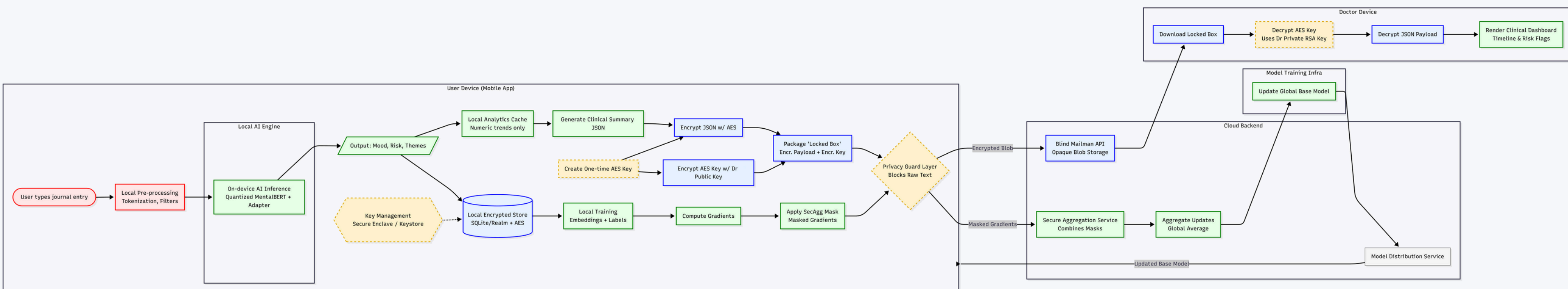
UNTRUSTED ZONE (BACKEND)

- **Coordinator Service:** A Python/FastAPI service responsible for routing encrypted "Locked Boxes" and coordinating the Federated Learning rounds.

CLINICIAN NODE

- **Decryption Client:** A local interface for the clinician that utilizes their private RSA key to decrypt the AES session keys contained in the "Locked Box," subsequently unlocking the patient's data.

System Architecture Overview



Workflow and Methodology

INGESTION PHASE

User input is immediately tokenized. A "Privacy Guard" layer acts as a firewall, blocking any HTTP request that contains raw text strings.

CARE LOOP (SECURE REPORTING)

1. The app generates a JSON summary of the user's trends.
2. A one-time AES-256 session key is generated to encrypt this JSON.
3. The session key is encrypted using the Doctor's Certified Public RSA Key.
4. The encrypted payload is uploaded to the backend.

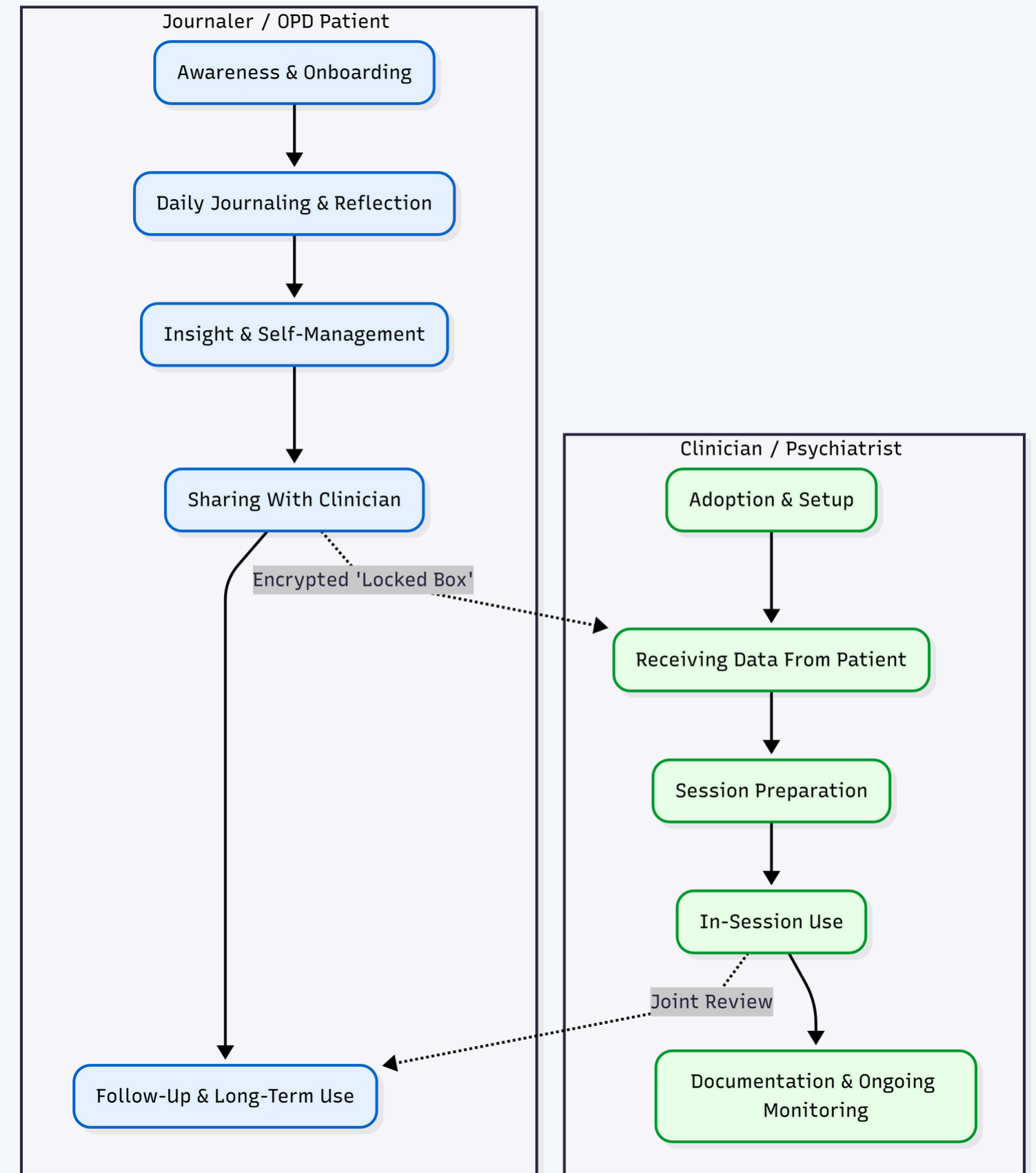
INFERENCE PHASE

The local MentalBERT model analyzes the tokens to generate a mood vector and check for DSM-5 risk flags (e.g., Depression, Anxiety, Self-Harm).

LEARNING LOOP (FEDERATED LEARNING)

1. During idle/charging states, the device retrains the model on local data.
2. Computed gradients are masked using the SecAgg protocol.
3. Masked vectors are sent to the aggregator.

Workflow and Methodology





Technology Stack Overview

FRONTEND: FLUTTER (DART)

Chosen for its ability to compile to native ARM code, ensuring high performance for the TFLite inference engine.

BACKEND: PYTHON 3.9+ / FASTAPI

Selected for its high-throughput asynchronous capabilities, essential for handling concurrent encrypted data streams.

DATABASE

- **Local:** Hive with SQLCipher (AES-256) extension.
- **Server:** PostgreSQL (Metadata) and Redis (Message Queue for Federated Learning coordination).

AI FRAMEWORKS

- **TensorFlow Lite** (for edge inference)
- **TensorFlow Federated** (for distributed training logic).



Module Breakdown

MODULE 1: SECURE JOURNALING SYSTEM

This module manages the user interface for data entry and handles the cryptographic operations (AES-256-GCM) required to persist data securely to the local filesystem.

MODULE 2: ON-DEVICE AI ENGINE

This background service manages the lifecycle of the TFLite interpreter. It handles text preprocessing (tokenization), model execution, and the post-processing of output logits into interpretable mood scores and risk flags.

MODULE 3: CLINICIAN "LOCKED BOX"

This security module implements the RSA-AES hybrid encryption protocol. It manages the storage of the clinician's public key and ensures that the ephemeral AES keys used for report encryption are securely discarded after use.



Feasibility

TECHNICAL FEASIBILITY

The primary challenge of running BERT on mobile is model size. We achieve feasibility through INT8 Quantization, reducing the model footprint to <50MB and inference latency to <200ms, well within the capabilities of mid-range smartphones.

ECONOMIC/RESOURCE FEASIBILITY

The "Local-First" design significantly lowers operational costs by offloading the computationally expensive inference tasks to client devices. This removes the need for expensive cloud GPU clusters.

DEVELOPMENT FEASIBILITY

The project follows a Spiral Model, which mitigates risk by validating high-risk components (encryption overhead, model accuracy) in early spirals before full integration.



Conclusion

TECHNICAL VALIDATION

Anora serves as a proof-of-concept that strict privacy requirements (Zero-Knowledge) and advanced functionality (AI Analysis) are not mutually exclusive.

SOCIETAL IMPACT

By optimizing for mid-range hardware and offline use, Anora democratizes access to advanced mental health monitoring. It provides a safe harbor for vulnerable users, ensuring that their mental health data cannot be subpoenaed or leaked, effectively treating privacy as a human right.

FUTURE DIRECTION

The architecture sets a new ethical standard for "Clean AI," proving that effective machine learning systems can be built without the invasive harvesting of personal data.

References

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3. *Model Source: "MentalBERT: Publicly Available Pre-trained Models for Mental Healthcare" (HuggingFace).*
4. *Clinical Guidelines: DSM-5 (Diagnostic and Statistical Manual of Mental Disorders) used for risk flag alignment.*